

Introduction to Linear Programming

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Introduction to Linear Programming

Linear Programming (LP): A mathematical optimization technique for achieving the best outcome in a mathematical model whose requirements are represented by linear relationships.

Basic Concepts

- **Decision Variables:** Parameters we can control or adjust.
- **Objective Function:** The goal expressed in terms of decision variables.
- **Constraints:** Limitations or conditions that must be satisfied.
- **Feasible Region:** Set of valid solutions within the solution space.
- **Optimization:** Process of finding the best solution.

Decision Variables

- **Definition:**

Decision variables are the parameters under our control, representing quantities to be determined.

- **Example:**

In a production scenario, x and y might represent the quantities of two different products to manufacture.

Objective Function

- **Definition:**

The objective function quantifies the goal we want to achieve, whether to maximize or minimize.

- **Example:**

Maximize profit ($P=3x+2y$).

Constraints

- **Definition:**

Constraints are conditions or limitations that restrict the possible values of decision variables.

- **Example:**

Limited resources: $2x+3y \leq 150$ (Machine hours constraint).

Feasible Region

- **Definition:**

The feasible region is where valid solutions, satisfying all constraints, exist within the solution space.

- **Example:**

Graphical representation on a coordinate plane.

Optimization

- **Definition:**

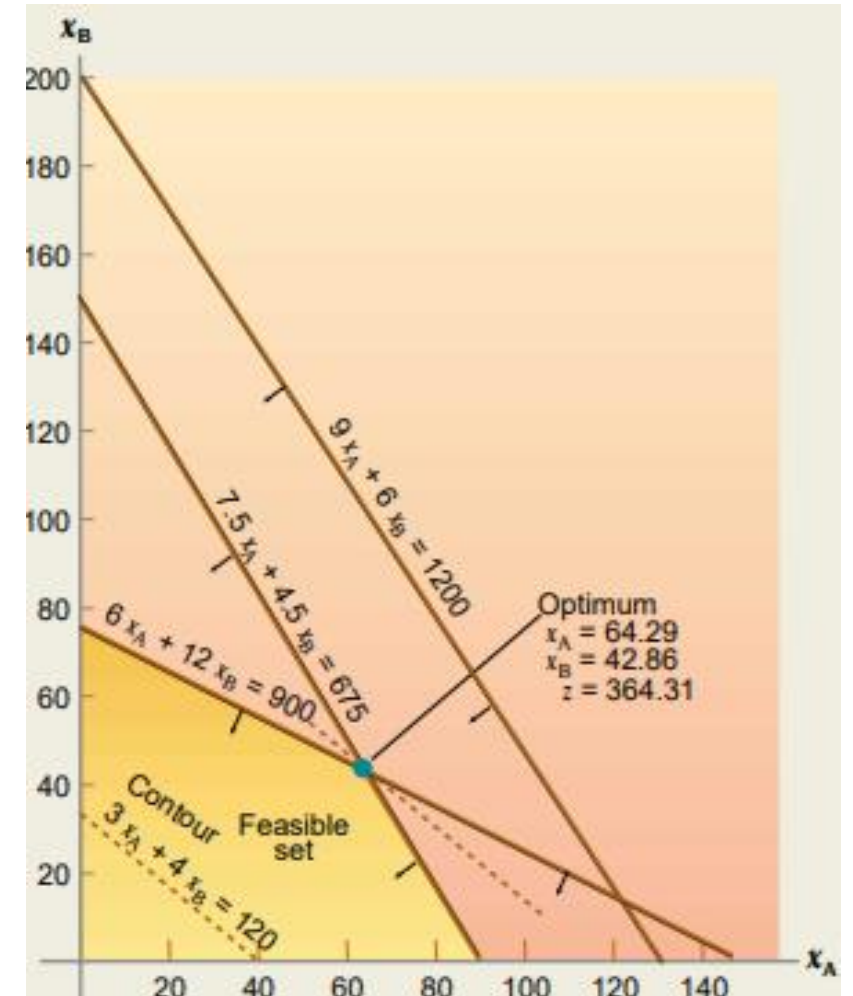
Optimization is the process of finding the best solution that either maximizes or minimizes the objective function.

- **Example:**

Balancing profit maximization with resource constraints.

Graphical Representation

Visualizing Linear Programming Problems



Setting Up the Coordinate Plane

1. Identify decision variables.
2. Assign axes to decision variables.
3. Plot constraints as lines or inequalities.

Example Graphical Representation

- **Example:**

$2x+3y \leq 150$ (Machine hours constraint).

$x, y \geq 0$.

Shading the Feasible Region

Steps:

1. Test a point within the feasible region.
2. Shade the side where the test point satisfies all inequalities.

Identifying Optimal Solution

- **Definition:**

The optimal solution lies at the vertex of the feasible region.

- **Example:**

Find coordinates of vertex points and evaluate the objective function at each.